MAJOR PROJECT PROTOTYPE V3

CREDITZ

AICTE 200 Activity Points & Degree Clearance System

An intelligent institutional platform streamlining co-curricular credit tracking, multimodal AI certificate verification, deterministic policy ceilings, and official NAAC/NBA accreditation clearance transcripts.

Version 3.0 Refactor Highlights:

- 1-Click Institutional PDF Credit Report (A4 Certified)
- Faculty Mentor Visual Analytics (Category Donut & Velocity)
- Bell Icon Real-time Meeting & Advisory Notice Space
- Minimalized, Non-IT Friendly Academic Vocabulary

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Autonomous Institute of Technology & Sciences | Evaluated for Major Project v3



CREDITZ PROTOTYPE V3

Addressing Evaluation Panel Feedback: The Evolution to v3

ITEM 01

Panel Feedback: Technical Jargon

Challenge:
Panel noted v2 had developer-centric terminology (Merkle root, cryptographic audit, schema hash) confusing non-IT academic panelists.

✓ V3 Resolution: Institutional Vocabulary

Completely converted to standard academic terms: 'Official PDF Transcript', 'Submissions', 'Digitally Verified ✓', and 'Point Rules & Categories'.

ITEM 02

Panel Feedback: Dashboard Clutter

Challenge:
Student dashboard had 8 mini-cards, duplicated upcoming events, and nested scroll traps making the page hard to navigate.

✓ V3 Resolution: Minimalized Current Focus

Unified 80% of screen to Current Semester focus card with clean expandable toggle, natural window scrolling, and unified Clubs/Events portal.

ITEM 03

Panel Feedback: Misplaced Visual Analytics

Challenge:
Complex velocity charts and donuts on student views caused confusion rather than providing clear clearance steps.

✓ V3 Resolution: Mentor-Exclusive Visuals

Removed visual charts from students and isolated them into a dedicated Faculty Mentor Analytics Desk (Donut, Velocity vs 30-pt cap, Trajectory).

ITEM 04

Panel Feedback: Accreditation & Communication

Challenge:
No printable physical document for NAAC/NBA inspections and no channel for quick mentor-student meeting updates.

✓ V3 Resolution: PDF Transcript + Bell Space

Engineered 1-click A4 Printable Transcript with QR verification hash and real-time Meeting Notice space in navbar Bell Icon.

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Project Objectives & Institutional Scope (Creditz v3)

[ROLE-BASED GOVERNANCE] 1. Multi-Role Institutional Architecture

Provide dedicated, tailored portals for Students, Faculty Mentors, and Head of Department (HOD) with strict Role-Based Access Control (RBAC) ensuring appropriate data privacy and workflow separation.

[MULTIMODAL AI OCR] 2. Intelligent Certificate Verification & Discrepancy Checks

Automate certificate verification using Google Gemini 1.5 Flash Vision OCR to extract recipient name, issuing agency, course, and date, computing an Authenticity Score (0-100%) against student records.

[POLICY ENGINE] 3. Deterministic Credit Ceiling & Domain Caps

Enforce AICTE continuous development rules mathematically: strict 30 points/semester ceiling (excess recorded as capped) and category maximums (e.g. MOOCs max 50 pts) to eliminate single-category credit farming.

[FACULTY ADVISORY] 4. Faculty Mentoring Desk & Targeted Visual Analytics

Empower faculty mentors with side-by-side certificate inspection, 1-click approvals, and exclusive visual analytics (category balance donut, velocity threshold chart, 200-pt degree trajectory curve).

[ACCREDITATION & COMM] 5. Official PDF Accreditation Transcript & Real-Time Sync

Deliver 1-click exportable A4 degree clearance transcripts with tamper-evident QR verification hashes and formal sign-offs for NAAC/NBA accreditation, plus real-time meeting notifications in the navbar bell icon.

● CREDITZ PROTOTYPE V3

System Architecture: 3-Tier Layered Design

ARCHITECTURE SPECIFICATIONS

● Presentation Layer:

React 19 Single Page App with TypeScript, Tailwind CSS v4, Lucide icons, and window-level vertical scrolling with responsive desktop/mobile navigation drawers.

● Application Backend:

Node.js Express REST API coordinating JWT authentication, Multer binary multipart uploads, real-time notification endpoints, and CORS security.

● Deterministic Rules Engine:

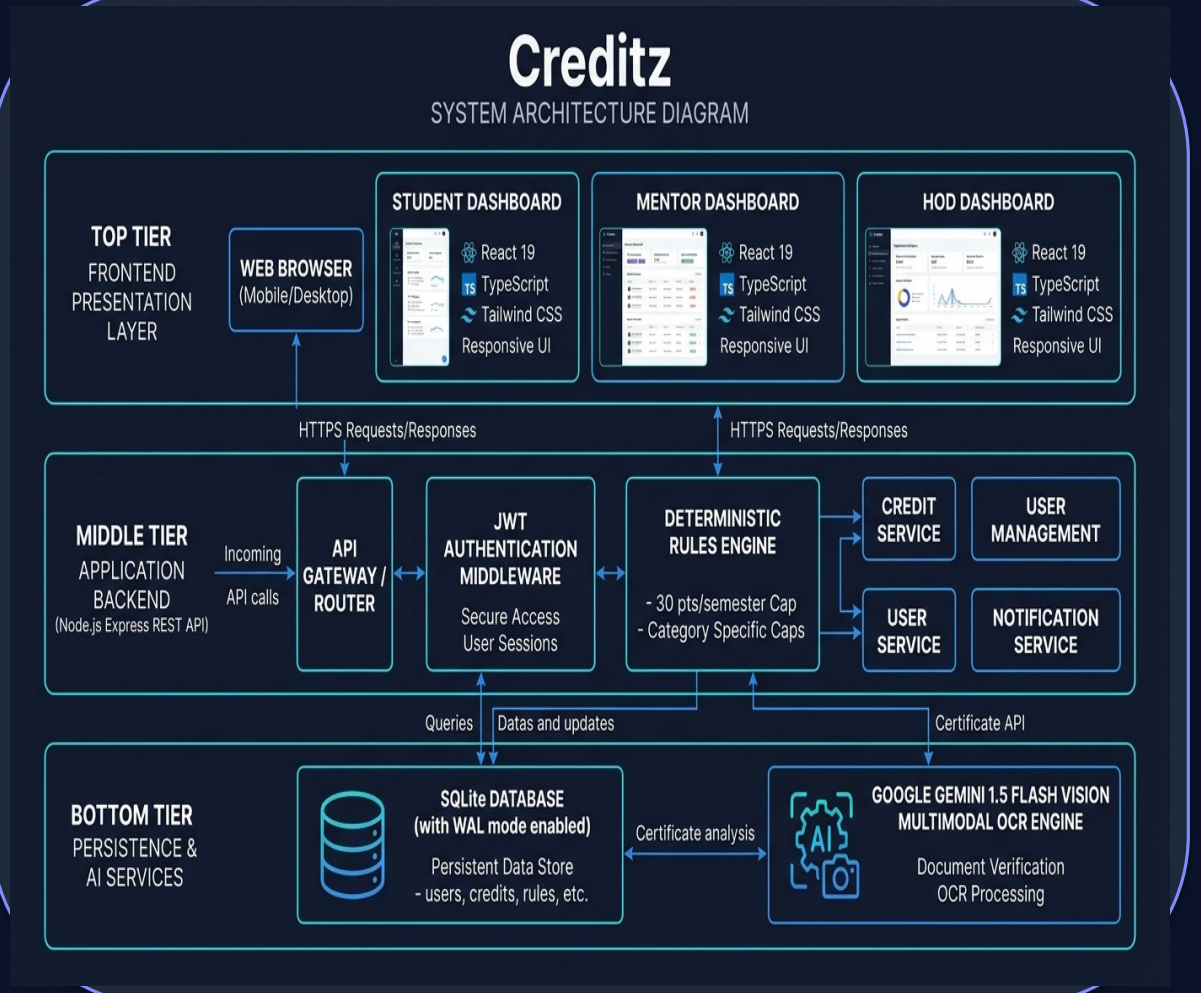
Independent mathematical module strictly enforcing the 30 pts/semester continuous development ceiling and category-specific maximum caps.

● AI Multimodal Vision Engine:

Google Gemini 1.5 Flash Vision via official SDK analyzing certificate buffers with smart deterministic heuristic fallback.

● Persistence Layer:

Relational SQLite database with Write-Ahead Logging (WAL) mode ensuring ACID transactional integrity and zero setup friction.



● CREDITZ PROTOTYPE V3

Relational Database Schema & Entity-Relationship Model

CORE RELATIONAL ENTITIES

● Users Entity:

Central identity management storing Student, Mentor, and HOD records with department FK, assigned mentor FK, USN roll number, CGPA, and semester.

● Submissions Entity:

Tracks student certificate uploads, claimed category, file URL, status ('pending', 'approved', 'rejected'), awarded points, semester, and AI audit JSON.

● Activity Schema & Categories:

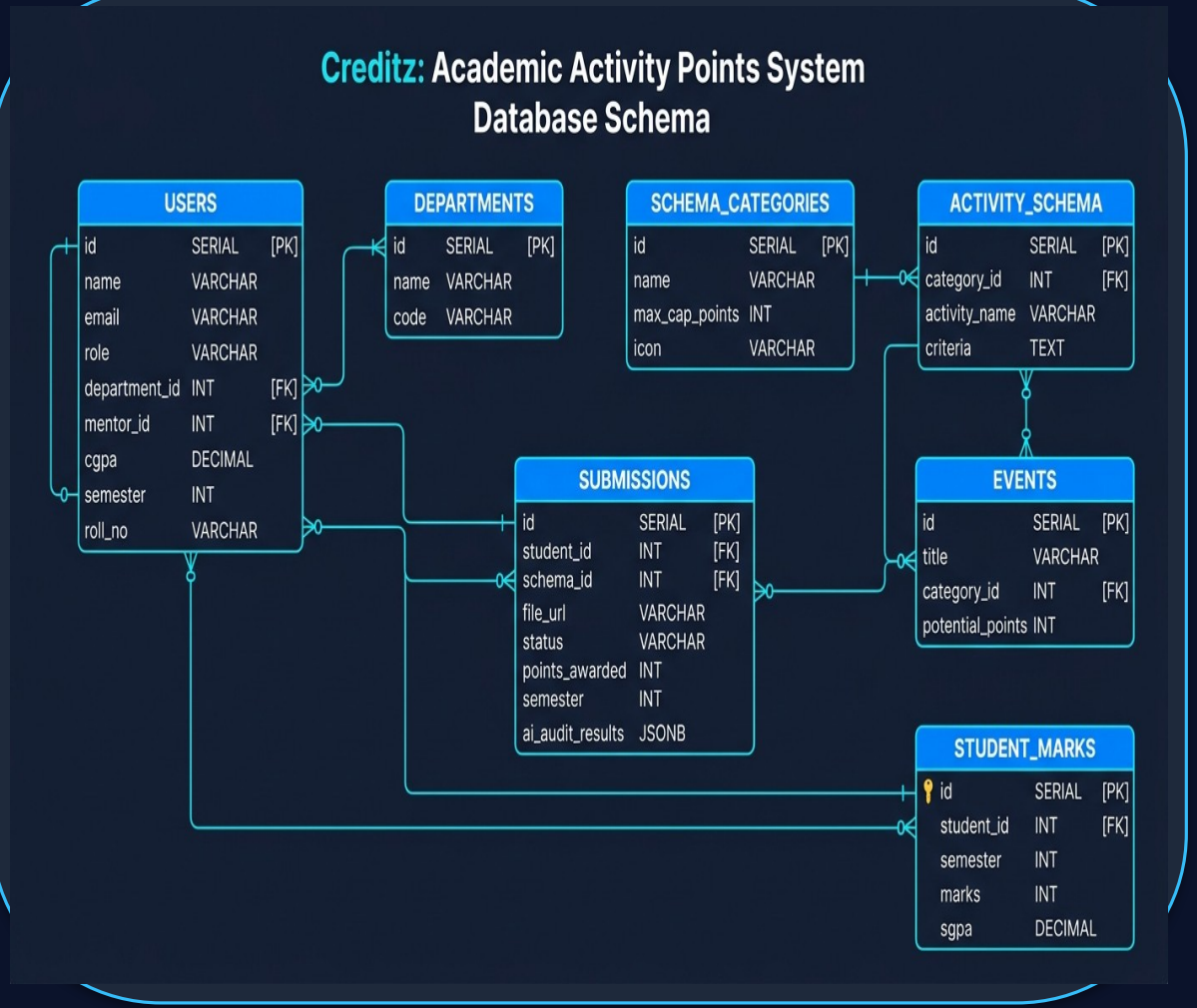
Institutional catalog of recognized activities, default points, verification criteria, and category domain maximum caps (e.g. max 60 pts technical certs).

● Notifications Entity:

Stores real-time institutional notices, sender/recipient IDs, meeting times, venues, and read status for seamless HOD-Mentor-Student communication.

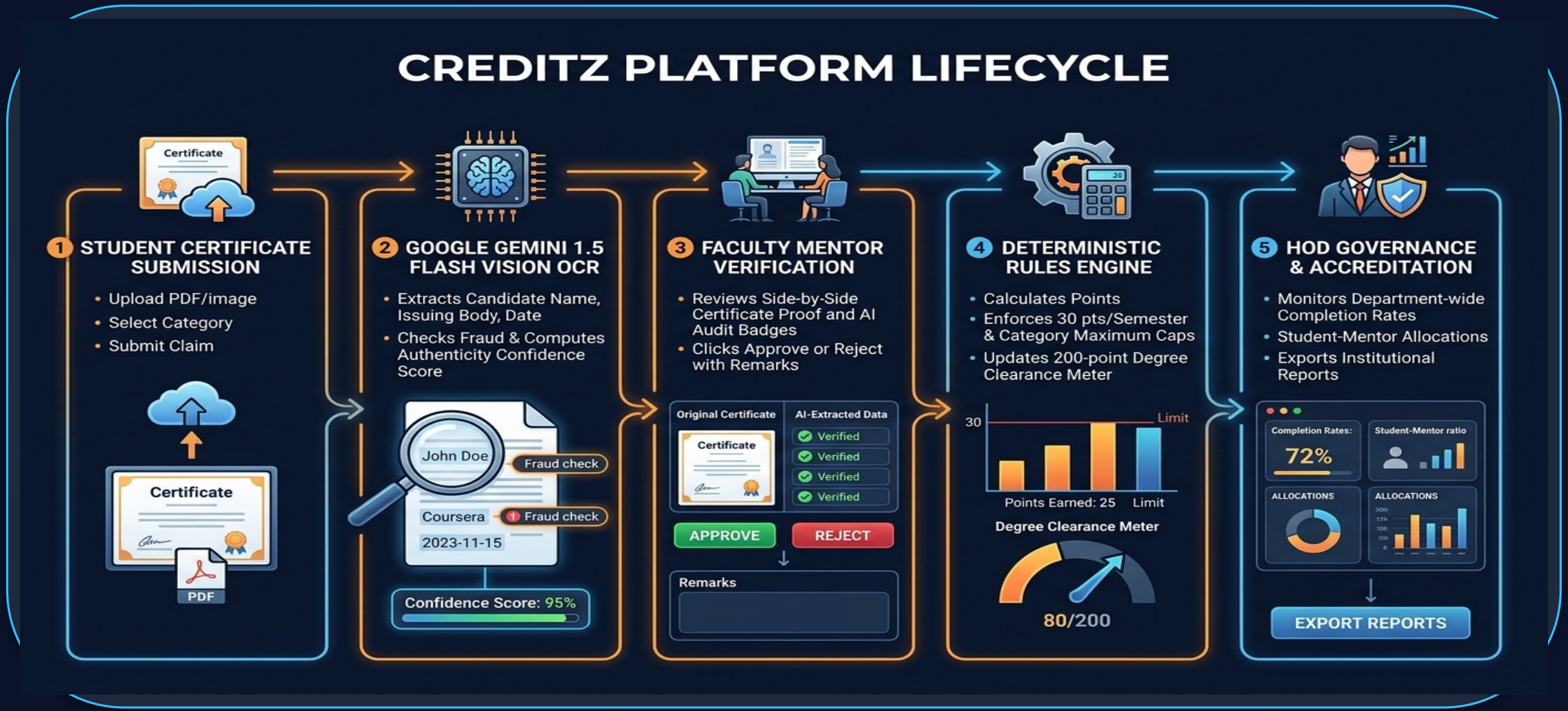
● Student Marks Entity:

Stores semester-wise CIE internal exam scores, lab marks, and SGPA calculations for holistic mentor academic guidance.



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End-to-End Operational Lifecycle & Verification Flow



CREDITZ PROTOTYPE V3

Multimodal AI Vision Inspection & Deterministic Rules Engine

VERIFICATION MECHANICS

✓ **Multimodal OCR Scanning:**
Google Gemini 1.5 Flash Vision reads binary document buffers (PDF, PNG, JPG), extracting student name, course title, issuing body, and completion date.

✓ **Cross-Verification Discrepancy Check:**
Compares extracted text against database profile to flag mismatched recipient names, invalid completion dates, or suspicious template patterns.

✓ **Authenticity Confidence Score:**
Computes an automated authenticity metric (0-100%) and generates plain-English audit flags for mentor evaluation.

✓ **Deterministic Semester Ceiling:**
Strictly caps counted activity credits to 30 points per semester. Excess credits are safely recorded as capped to mandate steady learning.

✓ **Offline Heuristic Fallback:**
Internal smart vision heuristic ensures 100% platform uptime even when internet connection or external API tokens are unavailable.



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Academic Literature Survey Summary (Clean Baseline)

Mishra et al. (2026, IEEE Transactions) — AI-Enhanced Blockchain-Based Certificate Verification

Key Finding: Hybrid CNN-BiLSTM architecture achieving 98.3% accuracy in visual and semantic certificate forgery detection. | Creditz Alignment: Direct theoretical foundation for Creditz's multimodal OCR pipeline and cross-field discrepancy checking.

Nguyen et al. (2025, IEEE Access) — Knowledge Graph Framework for Degree Pathway Recommendations

Key Finding: Semantic representation of course prerequisites, curriculum milestones, and institutional constraint enforcement. | Creditz Alignment: Informed Creditz's domain category cap structure and semester credit velocity tracking.

Chen et al. (2018, IEEE Transactions on Software Engineering) — DRLGENCERT: Automated Testing of Certificate Verification

Key Finding: Reinforcement learning for certificate feature extraction, boundary case exploration, and discrepancy detection. | Creditz Alignment: Guided the automated validation rules for issuing body matching and date range checks.

Zhang et al. (2024, IEEE Transactions on Learning Technologies) — Temporal Point Process Model for Academic Progression

Key Finding: Models learning motivation cycles and temporal credit accumulation patterns across undergraduate programs. | Creditz Alignment: Empirical justification for enforcing the AICTE 30-point continuous development semester ceiling.

Edge et al. (2024, Microsoft Research) — From Local to Global: Graph RAG Summarization

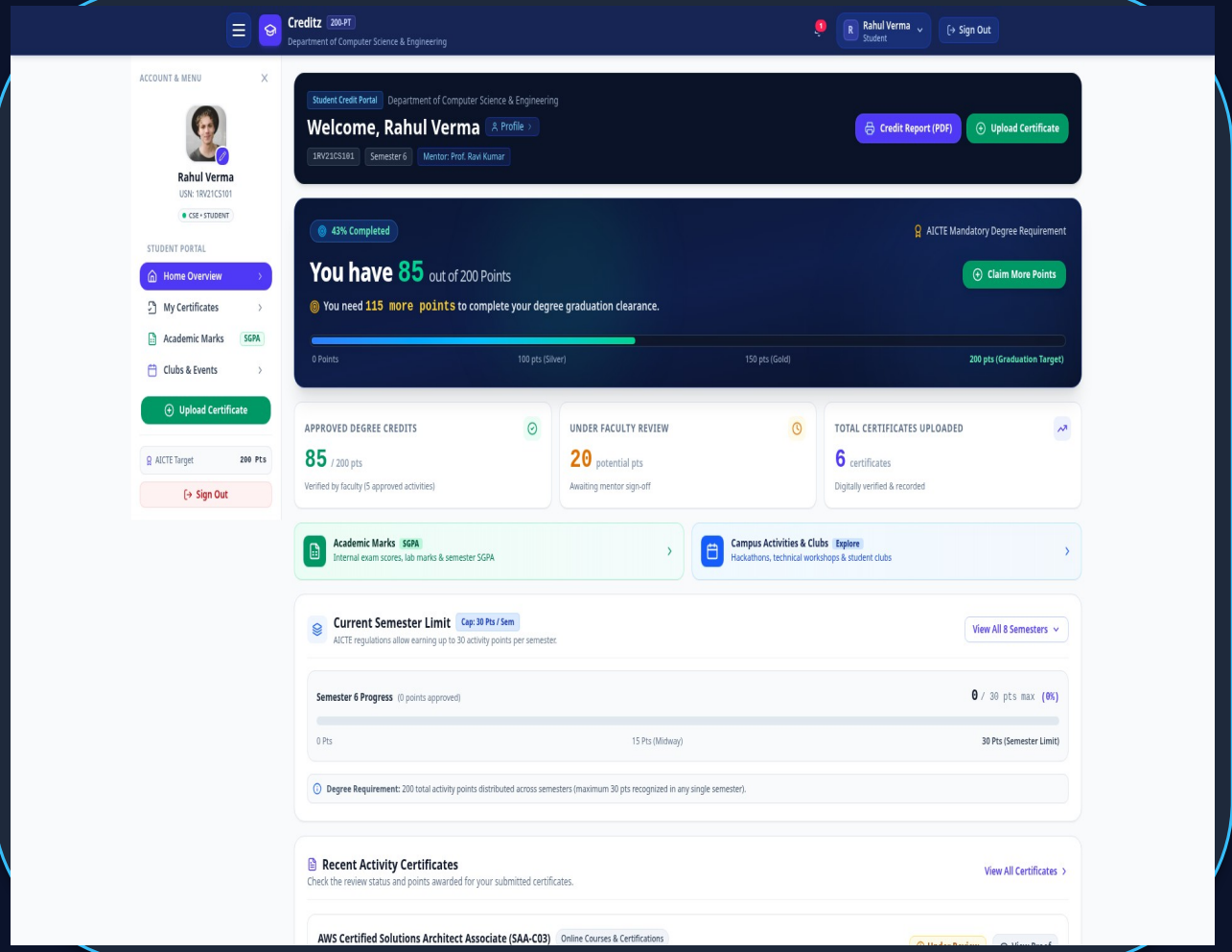
Key Finding: Demonstrated that grounded contextual graphs eliminate LLM hallucinations during document verification. | Creditz Alignment: Reinforces Creditz's deterministic schema grounding over raw generative AI outputs.

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Role 1: Student Degree Clearance Portal (Version 3 Minimalized UI)

STUDENT CAPABILITIES (V3)

- **200-Point Clearance Meter:**
  Live visual gauge tracking total approved credits toward graduation with milestone tiers (Bronze 50, Silver 100, Gold 150, Diamond 200).
- **1-Click Official PDF Transcript:**
  Prominent button opening the official A4 degree transcript modal with verification hash for instant print and download.
- **Current Semester Focus Card:**
  Minimalized card displaying current semester progress vs the 30 pts/sem limit, with expandable 8-semester historical drawer.
- **Plain-English Status Badges:**
  Replaced developer jargon with clear labels: 'Digitally Verified & Recorded', 'Under Faculty Review', 'Approved Degree Credits'.
- **Interactive Meeting Space:**
  Direct bell-icon popover to request advisory meetings with faculty mentor and view scheduled sessions.



CREDITZ PROTOTYPE V3

Institutional Innovation: 1-Click Official PDF Credit Report

ACCREDITATION REPORT SPECS

✓ **NBA & NAAC Institutional Letterhead:**

Formally formatted with institution title, accreditation status (A++ Grade), state university affiliation, and CSE department seal.

✓ **Unique Verification Hash & QR Code:**

Generates a unique tamper-evident verification reference string (CRD-CSE-1RV21CS101-PCW9QU) for instant physical document validation.

✓ **Continuous Learning Ceiling Breakdown:**

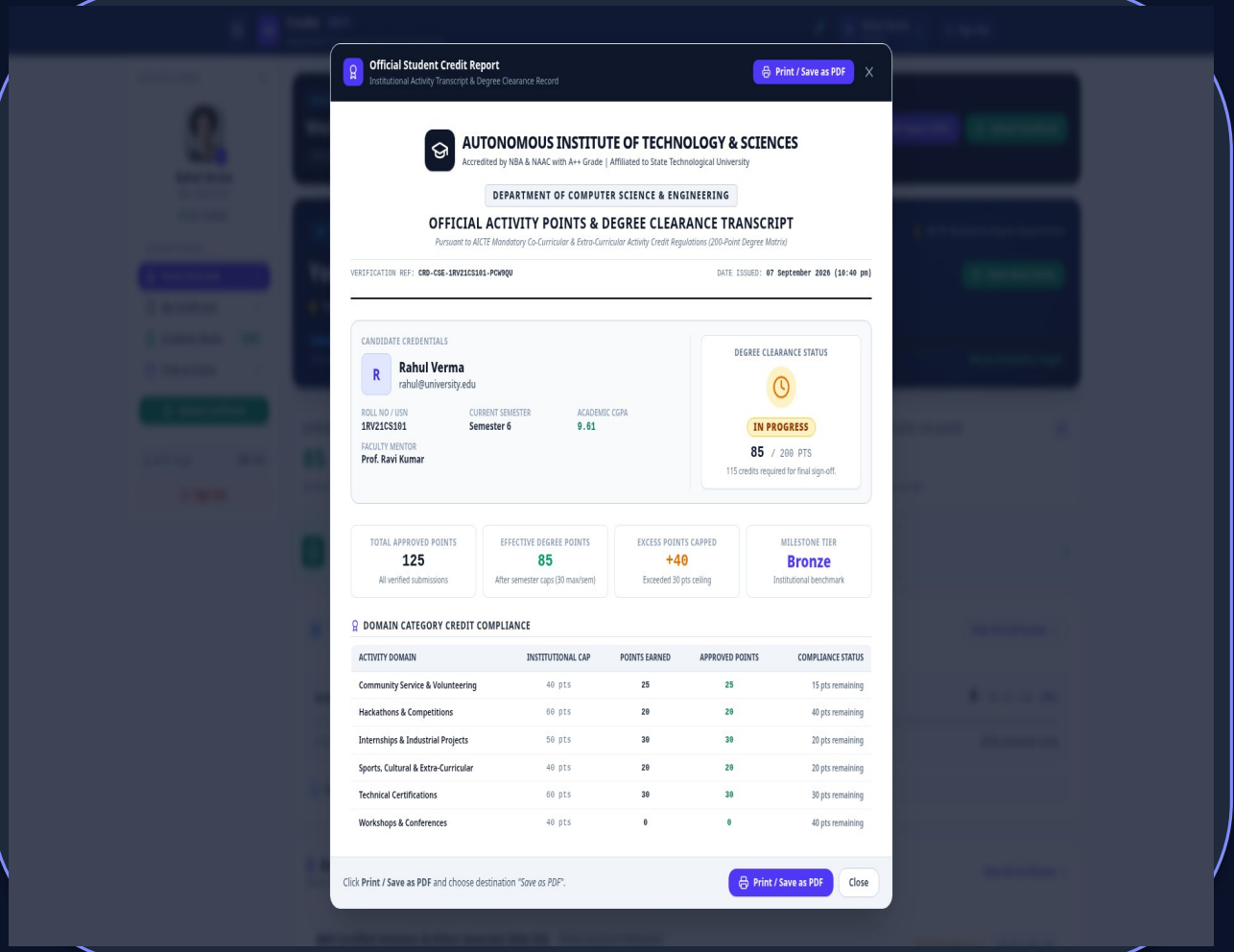
Differentiates Total Approved Points (125 pts) from Effective Degree Points (85 pts) and Capped Excess (+40 pts) per AICTE rules.

✓ **Category Domain Compliance Table:**

Verifies compliance against institutional caps across Community Service, Hackathons, Internships, and Technical Certifications.

✓ **Official Sign-off Section:**

Dual formal authorization blocks with designations for Faculty Mentor and Head of Department for graduation audit portfolios.

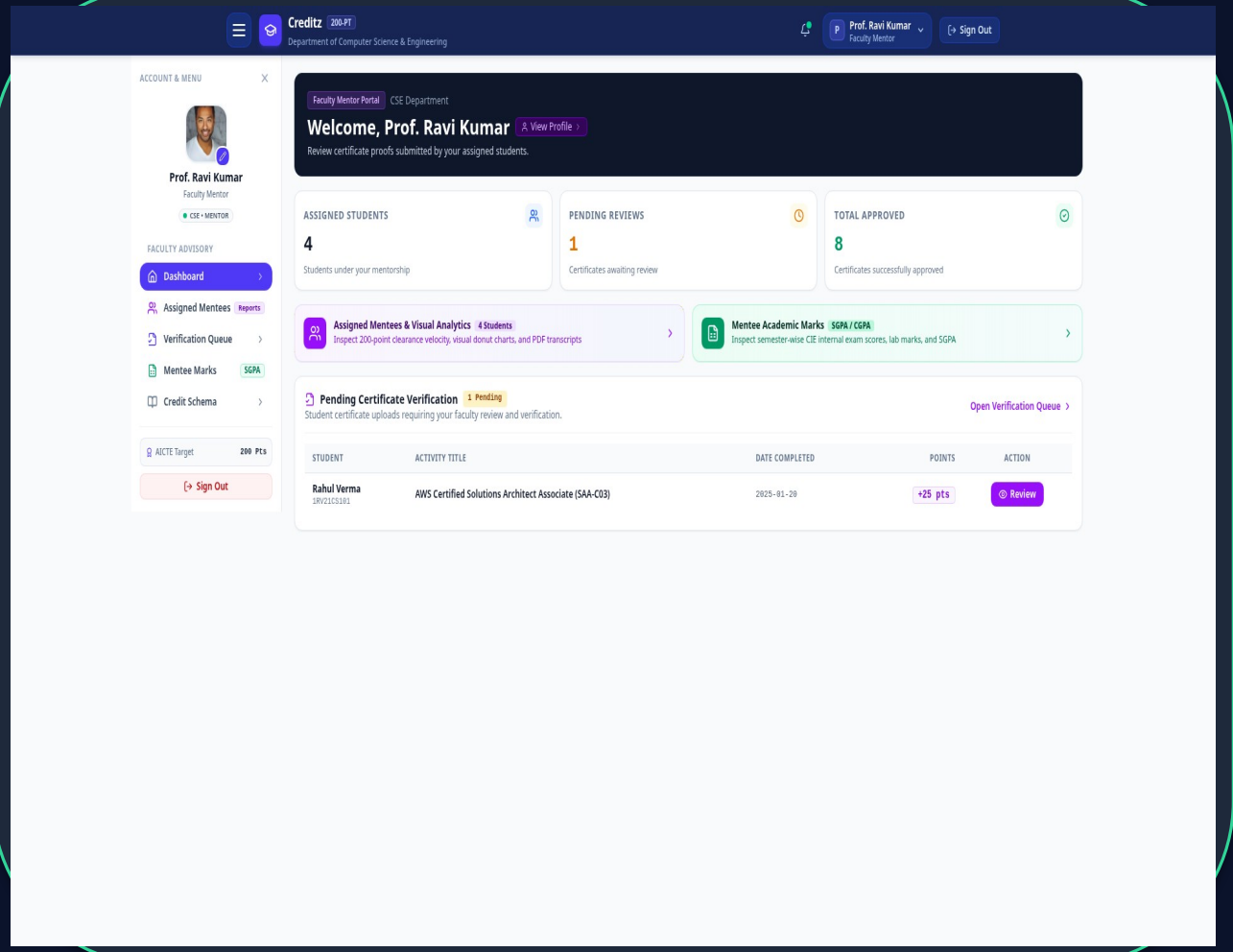


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Role 2: Faculty Mentor Advisory & Verification Console

MENTOR CONSOLE CAPABILITIES

- **Action-Oriented Dashboard:**
  Replaced crowded 10-student card grid with 2 clean shortcut cards (Assigned Mentees & Mentee Academic Marks) and prominent Pending Review alerts.
- **Pending Certificate Queue:**
  Instant visibility into unreviewed student submissions (e.g. AWS Cloud Certification) requiring faculty signature and points allocation.
- **Side-by-Side Proof Viewer:**
  Inspects uploaded certificate image/PDF alongside AI multimodal OCR extraction fields and authenticity confidence metrics.
- **1-Click Verification Decisions:**
  Approve, reject, or request revisions with structured faculty feedback recorded in student timeline.
- **Holistic Mentee Monitoring:**
  Track academic semester marks (CIE, labs, SGPA) alongside co-curricular points to ensure balanced student development.



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Faculty Mentor Visual Analytics (Targeted Mentorship Intelligence)

ANALYTICS DESK HIGHLIGHTS

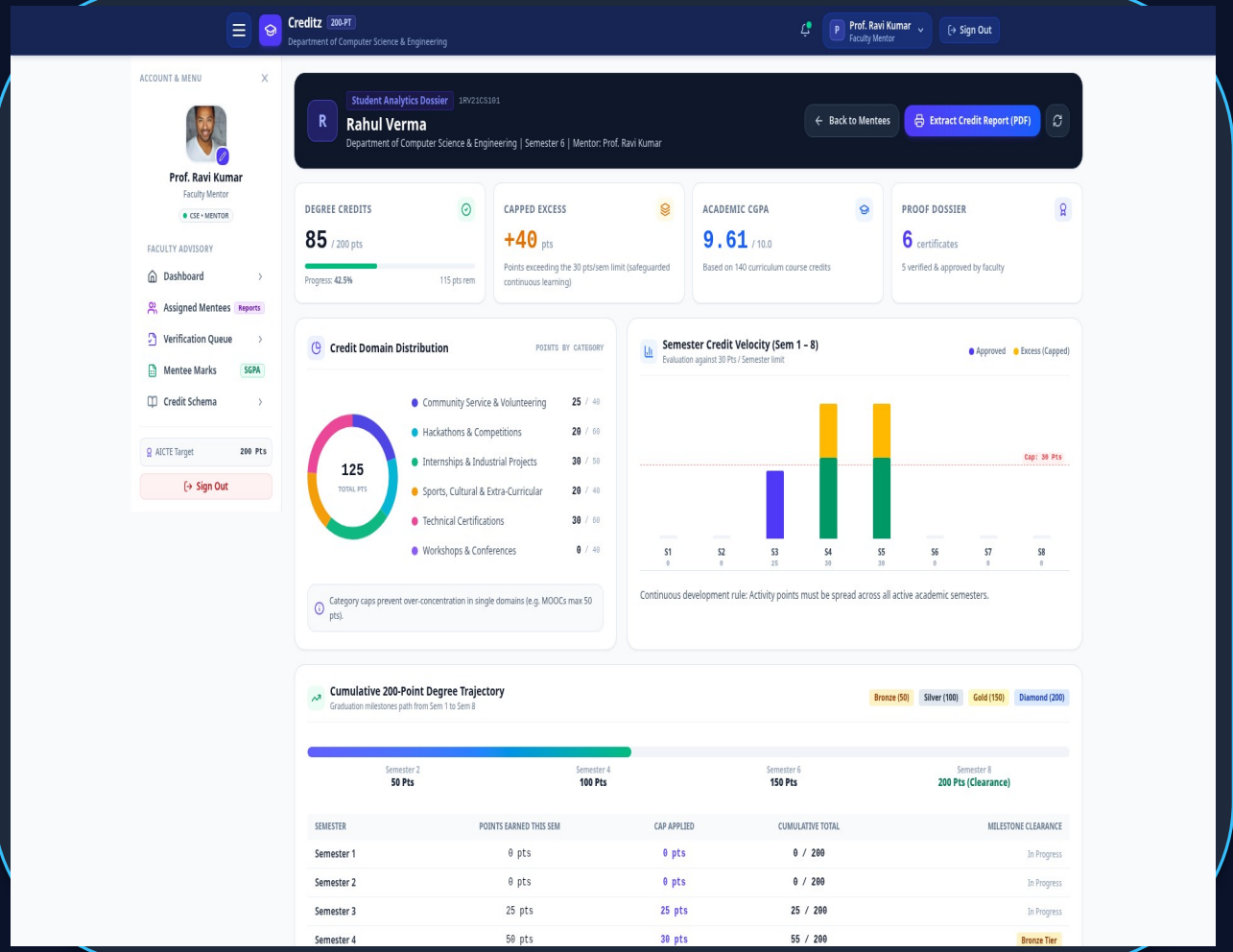
✓ **Isolated to Faculty Mentors:**
Visual charts intentionally removed from student views to avoid confusion, and consolidated into an in-depth Faculty Advisory workspace.

✓ **Category Domain Donut Chart:**
Visualizes mentee points breakdown across all 6 activity domains, immediately exposing category cap saturation or domain deficits.

✓ **Semester Credit Velocity Chart:**
Bar chart plotting semester-wise credits against the red dashed 30-point ceiling line, highlighting approved credits vs capped excess.

✓ **200-Point Degree Trajectory Curve:**
Milestone progress gauge (Bronze 50, Silver 100, Gold 150, Diamond 200) with detailed semester audit log table.

✓ **Proactive Academic Intervention:**
Enables mentors to identify students falling behind in activity credits before graduation year deadlines.

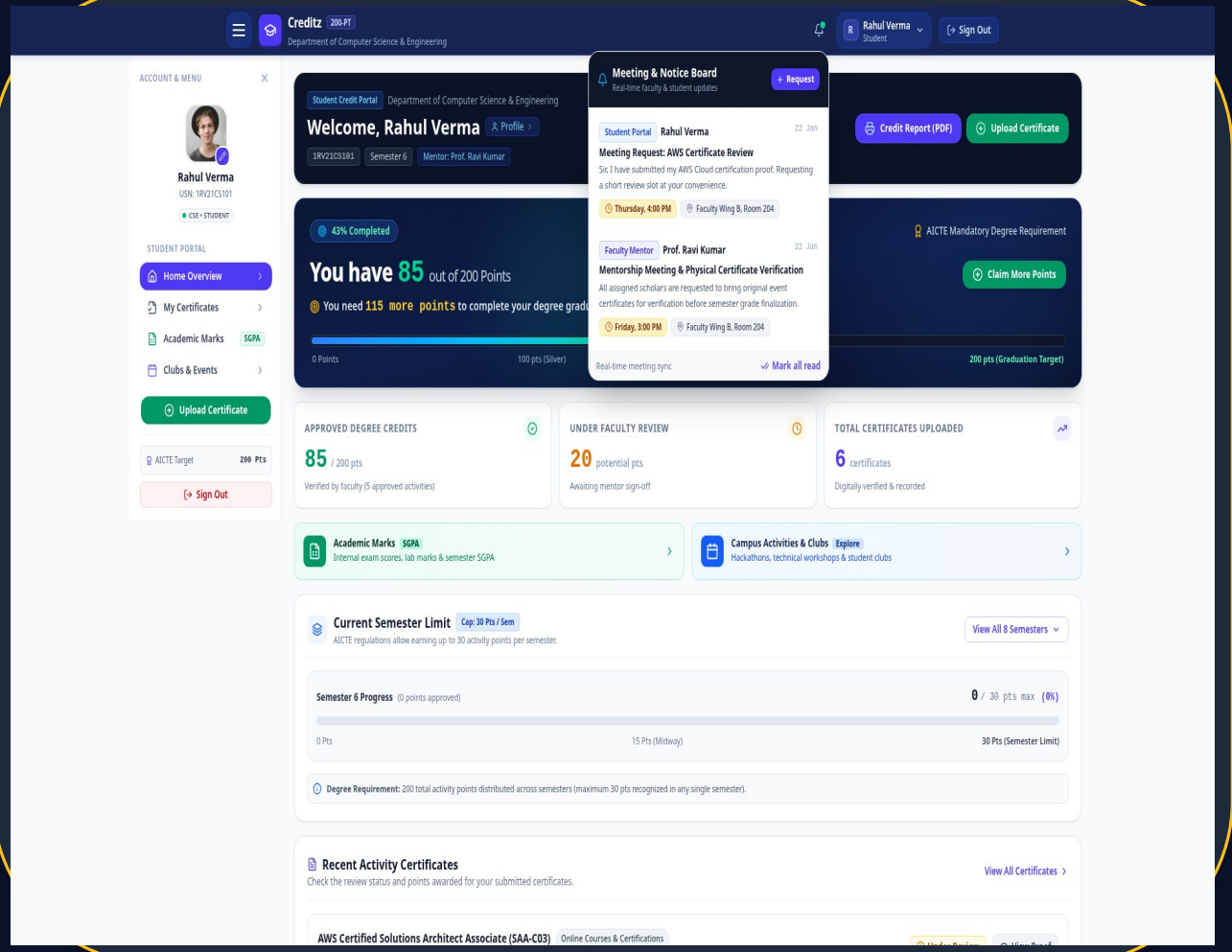


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Real-Time Meeting & Advisory Notice Space (Navbar Bell Icon)

COMMUNICATION HIGHLIGHTS

- **Zero-Clutter Navbar Integration:**
  Clean message space housed entirely within the navbar bell icon popover without cluttering dashboards or sidebar navigation.
- **HOD to Mentors Communication:**
  Head of Department can broadcast urgent circulars, accreditation deadlines, and scheduled faculty review meetings.
- **Mentor to Mentees Advisories:**
  Mentors schedule physical certificate verification meetings, specify room/venue (e.g. Faculty Wing B, Room 204), and set meeting times.
- **Student Meeting Requests:**
  Students can initiate meeting requests with their assigned faculty mentor directly from the popover with a 1-click form.
- **Real-Time Synchronization:**
  Dynamic unread badge counter (pulsing red alert) and real-time polling updates keep all parties aligned without page reloads.

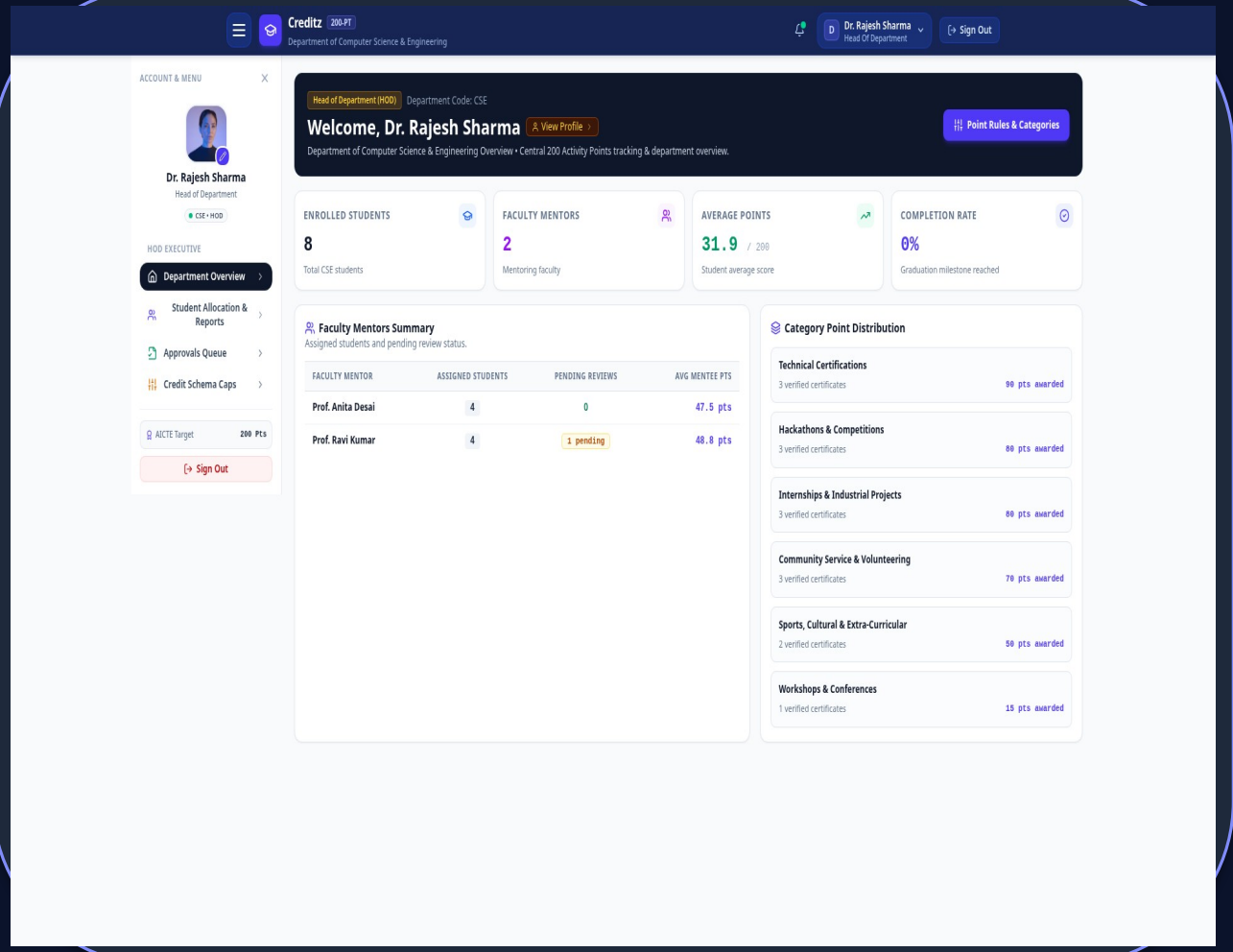


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Role 3: Head of Department (HOD) Governance & Oversight

HOD GOVERNANCE HIGHLIGHTS

- **Department Clearance Analytics:**
  Live department-level overview tracking total enrolled students (8), faculty mentors (2), average points (31.9/200), and graduation clearance rate.
- **Faculty Mentors Summary:**
  Real-time visibility into mentor workloads, mentee-to-faculty allocations, and pending review counts to prevent evaluation bottlenecks.
- **Category Point Distribution:**
  Aggregated analysis of points earned across Technical Certifications, Hackathons, Internships, Volunteering, and Sports domains.
- **De-jargonized Rule Management:**
  Simplified 'Point Rules & Categories' management console allowing HOD to adjust category maximum caps dynamically.
- **1-Click Accreditation Export:**
  Instant CSV and consolidated transcript exports for AICTE, NAAC, and NBA statutory compliance committee visits.



CREDITZ PROTOTYPE V3

Technology Stack & Version 3 Engineering Architecture

[FRONTEND PRESENTATION]

React 19 & Tailwind CSS v4

Stack: React 19, TypeScript, Vite 6, Tailwind CSS v4, Lucide Icons, Framer Motion

Responsive SPA with zero scroll traps; dynamic role-tailored dashboards, accessible modal dialogs, and native CSS print stylesheets for official PDF generation.

[APPLICATION RUNTIME]

Node.js 20/22 & Express REST API

Stack: Node.js (ESM), Express 4.21, TypeScript (tsx), JWT Authentication, Multer

High-throughput API server handling authenticated endpoints, binary file uploads, deterministic rule calculations, and real-time meeting notification polling.

[ARTIFICIAL INTELLIGENCE]

Multimodal Vision OCR Engine

Stack: Google Gemini 1.5 Flash Vision (@google/genai SDK) + Smart Local Fallback

High-speed multimodal document understanding model extracting candidate name, issuing body, and dates, paired with an offline heuristic fallback for 100% reliability.

[PERSISTENCE & RELIABILITY]

SQLite WAL Mode (Write-Ahead Logging)

Stack: better-sqlite3 v12.11 with WAL Mode & Immediate Transactions

ACID-compliant relational database with concurrent reader/writer support, sub-millisecond query latency, and zero-configuration single-file portability.

CREDITZ PROTOTYPE V3

Conclusion & Future Roadmap

KEY ACHIEVEMENTS IN PROTOTYPE V3

✓ **Full Alignment with Panel Feedback:**
Successfully purged cognitive clutter and replaced intimidating IT jargon with clear, non-IT friendly institutional language.

✓ **1-Click Official PDF Clearance Transcript:**
Engineered A4 printable accredited transcript with tamper-evident QR verification hash and faculty/HOD sign-off blocks.

✓ **Faculty-Exclusive Visual Analytics:**
Equipped mentors with multi-dimensional data analytics (category donut, velocity against 30-pt cap, degree trajectory) while simplifying student views.

✓ **Real-time Meeting & Notice Space:**
Added real-time notification popover to navbar bell icon enabling seamless HOD → Mentor → Student meeting coordination.

✓ **Deterministic AICTE Compliance:**
Mathematically guarantees continuous development via 30 pts/semester cap and category ceilings with Google Gemini 1.5 Flash Vision OCR.

FUTURE ROADMAP & PRODUCTION EXPANSION

National DigiLocker API Integration:
Direct integration with India's DigiLocker / NAD system for automated zero-upload certificate verification directly from issuers.

Blockchain Credential Anchoring:
Anchor finalized degree clearance hashes on Hyperledger or Ethereum testnet for tamper-proof post-graduation verification by employers.

Native Mobile PWA with Push Alerts:
Progressive Web App (PWA) with native mobile push notifications for real-time meeting reminders and event registrations.

Inter-College Activity Point Portability:
Standardized activity credit interchange format allowing seamless credit transfer for lateral entry and university transfer students.

Thank You!

CREDITZ (Prototype v3)

An Intelligent Activity Credit Management & Degree Clearance System

Open for Evaluation Panel Feedback, Demonstration & Discussion

Department of Computer Science & Engineering
Autonomous Institute of Technology & Sciences